# Notes: Crane, Michenaud, and Weston (RFS, 2016)

## The Effect of Institutional Ownership on Payout Policy: Evidence from Index Thresholds

## Contents

# 1 Big picture

## 1.1 Research question

The paper asks a clean causal question:

> Does higher institutional ownership *cause* firms to pay out more cash to shareholders (dividends and repurchases)?

A central challenge is endogeneity: institutional investors may *select* into firms with certain payout policies (tax clientele, governance preferences, information environment), and firms may adjust payout in anticipation of investor demand.

## 1.2 Main contribution (conceptual)

The paper's contribution is to use a transparent institutional feature—Russell index reconstitution—to build a strong instrument for institutional ownership. This delivers:

- a credible causal estimate of the effect of institutional ownership on payout policy;
- evidence that the mechanism is consistent with **agency-cost reduction / monitoring** rather than pure correlation;
- a connection between **passive-style ownership changes** (indexing and benchmarking) and **real corporate policy**.

# 2 Institutional setting: Russell 1000/2000 reconstitution

## 2.1 How the Russell indexes create a quasi-experiment

Each year Russell forms:

- **Russell 1000:** the 1,000 largest U.S. firms by market capitalization,
- **Russell 2000:** the next 2,000 firms.

A firm close to the cutoff can land either as the smallest firm in the Russell 1000 or the largest firm in the Russell 2000. Two forces generate a discontinuity in index-driven demand at the cutoff:

1. **Benchmarking intensity differs:** far more dollars track/benchmark to the Russell 2000 than to the marginal tail of the Russell 1000.
2. **Index weights jump:** the largest firms in the Russell 2000 receive large weights, while the smallest in the Russell 1000 receive tiny weights.

**Figure 1** visualizes the weight discontinuity that motivates the relevance of the instrument.

### 2.2 Important institutional complications (and the paper's fixes)

The paper stresses three practical issues and how to address them:

- **Float adjustment:** Russell adjusts market caps using a float methodology after assignment. The authors use May 31 unadjusted market cap ranks and add a control for the float adjustment.
- **Banding (post-2006):** Russell introduced "banding" rules that reduce turnover near the cutoff. The sample is restricted to years **through 2006**.
- **Comparability at the cutoff:** identification relies on local continuity; the paper works in narrow bandwidths around the threshold.

## 3 Data and variable construction

### 3.1 Sample

- Russell 1000 and Russell 2000 constituents from 1991–2006.
- Merged to Compustat/CRSP and 13F (Thomson) for institutional ownership.
- Governance engagement measures from ISS (shareholder proposals and proxy voting).

### 3.2 Key variables

**Institutional ownership.** Total institutional ownership (IO) is the fraction of shares held by 13F institutions. The paper also decomposes IO into Bushee-style types: **Dedicated**, **Quasi-indexers**, and **Transient** (Table 2).

**Payout policy.** Outcomes include dividends, dividend yield, probability of paying a dividend, total payout, and repurchases (Table 3). Dollar variables are in real terms (2005 dollars) and some outcomes are in logs.

**Proxy engagement.** Outcomes include counts of proposals (management vs shareholder, governance vs SRI), probability of at least one governance/SRI proposal, proxy voting participation, votes against, and probability of proposal passage (Table 4).

**Agency proxies.** Cross-sectional proxies include CEO-chair duality, CEO ownership, board size, GIM index, ROA, and a high cash-flow/low market-to-book indicator (Table 5).

## 4 Identification strategy and empirical models

### 4.1 Running variable and local design

Let $Rank^*_{i,t}$ be firm $i$'s May 31 market cap rank used for assignment. Define distance to the cutoff:

$$x_{i,t} \equiv Rank^*_{i,t} - 1000,$$

so $x_{i,t} = 0$ at the threshold (smallest Russell 1000 firm), $x_{i,t} > 0$ are slightly smaller firms (into the Russell 2000), and $x_{i,t} < 0$ are slightly larger firms (deeper in the Russell 1000).

Let $R2000_{i,t}$ be an indicator for inclusion in the Russell 2000 in year $t$.

### 4.2 Model 1: sharp RD first stage for IO (Equation (1))

The first stage is a sharp RD specification estimating the discontinuity in IO at the threshold:

$$IO_{i,t} = \alpha_t + \tau\, R2000_{i,t} + \delta_1 x_{i,t} + \delta_2 (R2000_{i,t} \times x_{i,t}) + \delta_3 FloatAdj_{i,t} + e_{i,t}. \tag{1}$$

**Interpretation:** $\tau$ measures the causal jump in institutional ownership induced by crossing the Russell cutoff, controlling for smooth size trends.

### 4.3 Model 2: payout policy second stage (Equation (2))

The causal effect of IO on payout is estimated by 2SLS:

$$Policy_{i,t+1} = \theta_t + \beta\, \widehat{IO}_{i,t} + \gamma_1 x_{i,t} + \gamma_2 (R2000_{i,t} \times x_{i,t}) + \gamma_3 FloatAdj_{i,t} + \eta_{i,t+1}, \tag{2}$$

where $\widehat{IO}_{i,t}$ is the fitted value from (1).

**Timing.** $IO_{i,t}$ is measured in the first quarter after index assignment (post-reconstitution). Policy outcomes are measured at the next fiscal year end.

**Estimation details.** Regressions include year fixed effects; standard errors are clustered by firm; the paper reports results for a **small bandwidth** (about $\pm 100$ ranks) and a **large bandwidth** (about $\pm 750$ ranks). Coefficients on $x_{i,t}$ are often reported per 100 ranks.

### 4.4 Model 3: governance engagement RD regressions (Table 4)

For governance engagement outcomes $G_{i,t}$ (proposals and voting), the paper estimates RD discontinuity regressions:

$$G_{i,t} = a_t + \rho\, R2000_{i,t} + b_1 x_{i,t} + b_2 (R2000_{i,t} \times x_{i,t}) + b_3 FloatAdj_{i,t} + u_{i,t}. \tag{3}$$

**Interpretation:** $\rho$ captures a discrete difference in engagement at the cutoff, consistent with changes in monitoring intensity.

### 4.5 Model 4: cross-sectional heterogeneity via agency proxies (Table 5)

To test an agency-cost mechanism, the paper interacts IO with an agency proxy $XS_{i,t}$:

$$\ln(Dividends)_{i,t+1} = c_t + \beta_1 \big(\widehat{IO}_{i,t} \times XS_{i,t}\big) + \beta_2 \widehat{IO}_{i,t} + \beta_3 XS_{i,t} + d_1 x_{i,t} + d_2 (R2000_{i,t} \times x_{i,t}) + d_3 FloatAdj_{i,t} + \varepsilon_{i,t+1}. \tag{4}$$

**Table 1**
**Summary statistics**

| *A. Russell 1000* | Mean | SD | p25 | Median | p75 |
|---|---|---|---|---|---|
| *Total institutional ownership* | 0.65 | 0.22 | 0.50 | 0.67 | 0.81 |
| *Dedicated ownership* | 0.32 | 0.18 | 0.19 | 0.29 | 0.43 |
| *Quasi indexer* | 0.15 | 0.12 | 0.06 | 0.12 | 0.21 |
| *Transient ownership* | 0.11 | 0.10 | 0.03 | 0.09 | 0.17 |
| *Dividends (M$ 2005)* | 119 | 201 | 0.00 | 34.59 | 127 |
| *Dividend yield* | 0.02 | 0.02 | 0.00 | 0.01 | 0.03 |
| *Repurchases (M$ 2005)* | 148 | 295 | 0.00 | 9.64 | 129 |
| *Payout (M$ 2005)* | 284 | 475 | 17 | 89 | 285 |
| *Total assets (B$ 2005)* | 1.06 | 1.61 | 0.15 | 0.39 | 1.12 |
| *Market value (B$ 2005)* | 7.29 | 11.40 | 1.55 | 3.00 | 6.99 |

| *B. Russell 2000* | Mean | SD | p25 | Median | p75 |
|---|---|---|---|---|---|
| *Total institutional ownership* | 0.52 | 0.28 | 0.28 | 0.50 | 0.73 |
| *Dedicated ownership* | 0.20 | 0.16 | 0.09 | 0.16 | 0.28 |
| *Quasi indexer* | 0.11 | 0.11 | 0.02 | 0.07 | 0.15 |
| *Transient ownership* | 0.10 | 0.11 | 0.02 | 0.07 | 0.14 |
| *Dividends (M$ 2005)* | 7.42 | 24 | 0.00 | 0.00 | 6.21 |
| *Dividend yield* | 0.01 | 0.02 | 0.00 | 0.00 | 0.02 |
| *Repurchases (M$ 2005)* | 7.83 | 28 | 0.00 | 0.00 | 2.65 |
| *Payout (M$ 2005)* | 15.20 | 41 | 0.00 | 2.19 | 14 |
| *Total assets (B$ 2005)* | 0.97 | 1.94 | 0.16 | 0.39 | 1.02 |
| *Market value (B$ 2005)* | 0.46 | 0.41 | 0.177 | 0.320 | 0.607 |

These tables present the summary statistics for firms that belong to the Russell 1000 index (panel A) and the Russell 2000 index (panel B). Variables are defined in Table A1.

**Interpretation:** if institutions mitigate agency problems, the marginal effect of IO on dividends should be larger when agency problems are more severe. In this specification:

$$\frac{\partial \ln(Dividends)}{\partial IO} = \begin{cases} \beta_2, & XS = 0, \\ \beta_2 + \beta_1, & XS = 1. \end{cases}$$

# 5 Descriptive evidence: summary statistics (Table 1)

## 5.1 What Table 1 shows

Table 1 reports summary statistics separately for Russell 1000 firms (Panel A) and Russell 2000 firms (Panel B).

**Key patterns.**

- **Institutional ownership is higher in the Russell 1000 on average** (mean about 0.65) than in the Russell 2000 (mean about 0.52), consistent with size/institutional preferences.
- **Payout levels differ strongly:** Russell 1000 firms have substantially higher average dividends and repurchases (in real dollars).
- These differences motivate the need for a **local design**: unconditional means conflate size with ownership.

# 6 Instrument relevance: IO discontinuity at the cutoff (Figures 1–2, Table 2)

## 6.1 Figure 1: index weights jump at the threshold

Figure 1 plots Russell index weights against rank. The key message is that the largest Russell 2000 firms have much larger index weights than the smallest Russell 1000 firms. This implies that crossing the threshold produces a discrete shift in index-driven demand, supporting instrument relevance.

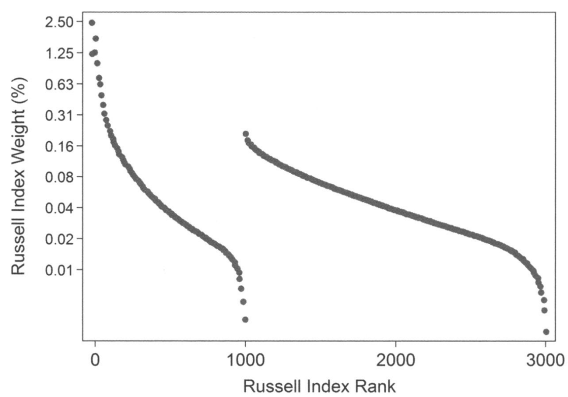


**Figure 1**
**Russell index weights around the threshold**
This figure shows the average index weights for firms in the Russell 1000 and the Russell 2000. Firms are assigned to the Russell 1000 or 2000 based on the firm's market capitalization at the end of May each year. Index weights are determined using a float-adjusted market capitalization within each index at the end of June.

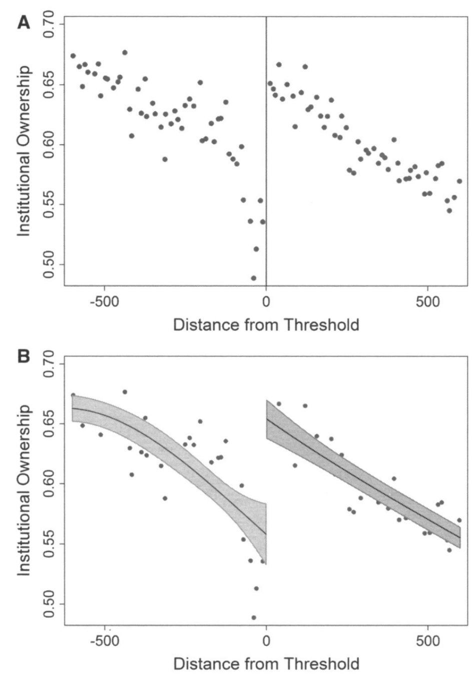


**Figure 2**
**Institutional ownership discontinuity**
This figure shows the total institutional ownership for the first quarter ending after the reconstitution of the Russell indexes for the Russell 3000 firms from 1991–2008. The *x*-axis represents the distance from the Russell 1000/2000 thresholds using the actual Russell ranks in the indexes, with zero representing the last firm in the Russell 1000. The figure plots the average total institutional ownership over ten ranks across all years (*A*) and adds regression discontinuity estimates and the associated 90% confidence bands following Equation (1) (*B*).

## 6.2 Figure 2: IO discontinuity is visible in the data

Figure 2 plots institutional ownership against distance from the threshold. The overall slope is negative (larger firms tend to have higher IO), but there is a visible **jump upward** at the cutoff for firms just inside the Russell 2000 relative to just inside the Russell 1000.

## 6.3 Table 2: quantifying the discontinuity (and who drives it)

Table 2 estimates (1) for total IO and for owner types.

**Panel A (small bandwidth).**

- Total IO jumps by about **0.09** (nine percentage points) for Russell 2000 firms relative to Russell 1000 firms at the cutoff.
- The jump is **not driven by dedicated owners** (near zero and insignificant).
- The jump is primarily driven by **quasi-indexers** and **transient owners** (both about 0.04 and statistically significant).

**Table 2**
**Differences in institutional ownership around the Russell 1000/2000 threshold**

| | (1) *Institutional ownership$_t$* | (2) *Dedicated ownership$_t$* | (3) *Quasi indexer$_t$* | (4) *Transient ownership$_t$* |
|---|---|---|---|---|
| *A. Small bandwidth* | | | | |
| $R2000_{it}$ | 0.09*** | 0.00 | 0.04*** | 0.04*** |
| | (4.20) | (0.16) | (3.07) | (3.14) |
| $(Rank^*_{it}\text{-}1000)$ | −0.08*** | 0.00 | −0.06*** | 0.00 |
| | (−2.82) | (0.32) | (−3.24) | (0.05) |
| $(Rank^*_{it}\text{-}1000) \times R2000_{it}$ | 0.12*** | −0.01 | 0.10*a* | 0.00 |
| | (3.60) | (−0.57) | (4.43) | (0.05 |
| $Float\ adj_{i,t}$ | 0.03*** | 0.01*** | 0.02*** | 0.01*** |
| | (17.18) | (7.29) | (11.27) | (9.84) |
| *Constant* | 0.36*** | 0.12*** | 0.17*** | 0.08*** |
| | (16.91) | (11.64) | (11.70) | (3.63) |
| Observations | 3,041 | 2,208 | 2,504 | 2,492 |
| R-squared | 0.22 | 0.10 | 0.14 | 0.11 |
| *B. Large bandwidth* | | | | |
| $R2000_{it}$ | 0.06*** | −0.00 | 0.04*** | 0.02*** |
| | (7.20) | (−0.72) | (7.59) | (4.29) |
| $(Rank^*_{it}\text{-}1000)$ | −0.01*** | −0.00 | −0.01*** | 0.00 |
| | (−7.94) | (1.01) | (−10.69) | (1.47) |
| $(Rank^*_{it}\text{-}1000) \times R2000_{it}$ | 0.00 | 0.00 | 0.01*** | −0.01*** |
| | (1.43) | (0.63) | (3.63) | (3.84) |
| $Float\ adj_{i,t}$ | 0.03*** | 0.01*** | 0.02*** | 0.00*** |
| | (24.70) | (3.87) | (27.17) | (21.20) |
| *Constant* | 0.40*** | 0.11*** | 0.20*** | 0.10*** |
| | (46.99) | (21.41) | (36.59) | (24.45) |
| Observations | 23,167 | 17,034 | 19,212 | 19,131 |
| R-squared | 0.27 | 0.10 | 0.13 | 0.33 |

**Table 3**
**Institutional ownership and payout: Instrumental variable estimates**

| *A. Small bandwidth* | (1) | (2) | (3) | (4) | (5) |
|---|---|---|---|---|---|
| *First stage* | IO | IO | IO | IO | IO |
| $\tau$ | 8.00*** | 8.19*** | 8.40*** | 9.71*** | 9.73*** |
| | (3.40) | (3.50) | (3.55) | (4.09) | (4.13) |
| *Second stage* | *Ln(Div.)* | *Div. Yield* | *Pr(Div.)* | *Ln(Total Pay)* | *Ln(Rep)* |
| IO | 6.57** | 0.10*** | 1.95* | 4.57* | 2.53 |
| | (1.98) | (2.30) | (1.83) | (1.93) | (1.31) |
| $(Rank^*_{it}\text{-}1000)$ | −0.22 | −0.00 | −0.16 | −0.15 | 0.05 |
| | (−0.92) | (−0.87) | (−1.25) | (−0.73) | (0.29) |
| $(Rank^*_{it}\text{-}1000) \times R2000_{it}$ | −0.14 | −0.00 | 0.12 | −0.24 | −0.40 |
| | (−0.26) | (−0.07) | (0.46) | (−0.53) | (−1.05) |
| *Float adj.* | −0.03 | −0.00 | 0.04 | 0.02 | 0.03 |
| | (−0.26) | (−1.36) | (0.60) | (0.22) | (0.34) |
| Year effects | y | y | y | y | y |
| Observations | 2,667 | 2,680 | 2,531 | 2,332 | 2,342 |
| *B. Large bandwidth* | | | | | |
| *First stage* | IO | IO | IO | IO | IO |
| $\tau$ | 4.34*** | 4.35*** | 4.38*** | 5.19*** | 5.19*** |
| | (5.10) | (5.14) | (7.48) | (6.04) | (6.06) |
| *Second stage* | *Ln(Div.)* | *Div. Yield* | *Pr(Div.)* | *Ln(Total Pay)* | *Ln(Rep)* |
| IO | 5.46*** | 0.05* | 0.92 | 4.39*** | 4.13*** |
| | (2.45) | (1.93) | (0.86) | (2.73) | (2.80) |
| $(Rank^*_{it}\text{-}1000)$ | −0.22*** | 0.00 | −0.06 | −0.28*** | −0.20*** |
| | (−9.72) | (0.24) | −(3.64) | (−17.54) | (−14.45) |
| $(Rank^*_{it}\text{-}1000) \times R2000_{it}$ | 0.17*** | 0.00 | 0.03 | 0.18*** | 0.15*** |
| | (6.12) | (0.27) | (2.32) | (7.86) | 7.13) |
| *Float adj.* | −0.07 | 0.00 | 0.03 | −0.03 | −0.09 |
| | (−0.94) | (0.03) | (0.81) | (−0.63) | (1.62) |
| Year effects | y | y | y | y | y |
| Observations | 20,614 | 20,740 | 19,486 | 18,007 | 18,056 |

**Panel B (large bandwidth).**

- The total IO jump remains positive (about 0.06).
- The quasi-indexer jump remains strong (about 0.04), and the transient component is smaller but significant (about 0.02).

**Interpretation.** The instrument mainly shifts ownership among index-like and benchmark-sensitive institutions, which is consistent with the Russell weight mechanism.

# 7 Main causal results: IO increases dividends and total payout (Table 3)

Table 3 reports 2SLS estimates using the RD discontinuity as the instrument for IO.

## 7.1 First stage in Table 3

The first-stage coefficient indicates that index assignment materially shifts IO. The reported magnitude corresponds to roughly **8–10 percentage points** in the small bandwidth and **4–5 percentage points** in the large bandwidth, consistent with Table 2 and Figure 2.

### 7.2 Second stage: payout effects

**Dividends.** Institutional ownership has a strong positive effect on dividends:

- ln(Dividends) rises significantly with instrumented IO in both bandwidths.
- The paper emphasizes the economic magnitude: a **one-percentage-point increase in IO causes roughly a $7 million (about 8%) increase in dividends**.
- Dividend yield increases and the probability of paying a dividend increases as well.

**Total payout and repurchases.**

- Total payout (dividends + repurchases) increases significantly with IO.
- Repurchases are positive but less stable in the small bandwidth; in the large bandwidth, repurchases become economically and statistically significant.

**Interpretation.** The causal estimates imply that institutional ownership pushes firms to distribute more cash to shareholders, with dividends as the most robust margin and repurchases as a complementary margin.

## 8 Mechanism evidence: proxy engagement and monitoring (Table 4)

The paper argues that institutions change payout partly by monitoring and governance engagement. Table 4 provides direct evidence from shareholder proposals and proxy voting outcomes around the cutoff.

### 8.1 Proposals

- There is **no meaningful discontinuity in management proposals**.
- There is a **large and significant increase in shareholder proposals** for Russell 2000 firms at the cutoff.
- The increase is concentrated in **governance-related proposals**, not in socially responsible investment (SRI) proposals.
- The probability of having at least one governance proposal increases at the cutoff.

### 8.2 Proxy voting behavior

- Votes *against* management increase for Russell 2000 firms, consistent with more active monitoring through voting.
- Voting participation and proposal passage show weaker discontinuities, but the overall pattern supports increased engagement where IO rises.

**Interpretation.** Even if many institutions are not "activists" in the classic sense, the proxy evidence suggests that increased institutional ownership is accompanied by governance involvement, consistent with an agency-based mechanism linking ownership to payout.

**Table 4**
**The difference in shareholder proposals and voting outcomes around the Russell 1000/2000 threshold**

| | Proposals | | | | | | Proxy voting | | |
|---|---|---|---|---|---|---|---|---|---|
| | Mgmt. | Shareholder | Gov. | SRI | Pr(gov) | Pr(SRI) | Participation | Votes against (%) | Pr(pass) |
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) |
| *$R2000_{it}$* | 0.03 | 0.63** | 0.73** | 0.33 | 0.09** | 0.06 | 0.035 | 0.07** | −0.10 |
| | (0.39) | (2.14) | (2.44) | (1.43) | (2.15) | (1.2) | (0.92) | (2.07) | (−1.52) |
| *($Rank^*_{it}$-1000)* | 0.01 | −0.05* | −0.05** | −0.21*** | −0.01*** | −0.05*** | 0.01*** | −0.01*** | 0.03*** |
| | (1.19) | (−1.95) | (−1.97) | (−8.59) | (−2.72) | (−9.29) | (3.12) | (−3.40) | (4.25) |
| ($Rank^*_{it}$ − 1000) × $R2000_{it}$ | −0.02 | 0.09** | 0.09** | 0.20*** | $0.01^b$ | 0.05*** | −0.02*** | 0.01** | −0.03*** |
| | (−1.42) | (2.35) | (2.28) | (6.11) | (2.26) | (6.23) | (−3.03) | (2.43) | (−3.62) |
| *Float $adj_{i,t}$* | 0.01 | 0.16** | 0.20*** | 0.04 | $0.02^b$ | −0.01 | 0.02*** | −0.01 | 0.02 |
| | (0.67) | (2.46) | (2.9) | (0.58) | (2.05) | (−0.58) | (3.64) | (−1.30) | (1.32) |
| *Constant* | 0.48*** | 3.98*** | 2.67*** | 0.61*** | 0.70*** | 0.20*** | 0.32*** | 0.34*** | 0.80*** |
| | (8.9) | (22.31) | (14.09) | (3.63) | (22.12) | (5.37) | (15.58) | (14.11) | (19.51) |
| Observations | 1,029 | 3,137 | 3,137 | 3,137 | 3,137 | 3,137 | 2,067 | 1,988 | 1,988 |
| R-squared | 0.003 | 0.010 | 0.010 | 0.065 | 0.008 | 0.104 | 0.020 | 0.012 | 0.016 |

**Table 5**
**Institutional ownership and dividends: Cross-sectional effects**

| | Agency measures (XS) | | | | | |
|---|---|---|---|---|---|---|
| | *CEO/ Chairman* | *CEO ownership* | *Board size* | *GIM index* | *ROA* | *High CF/ Low MTB* |
| *$IO_{i,t}XS_{i,t}$* | 3.77** | −1.47*** | 7.70* | 8.18*** | −3.62*** | 5.40*** |
| | (2.25) | (−7.08) | (1.67) | (2.99) | (−3.05) | (2.76) |
| *$IO_{i,t}$* | 2.73 | 3.68 | 6.78** | 2.79 | 7.35*** | 1.34 |
| | (1.07) | (1.24) | (1.97) | (0.89) | (2.70) | (0.87) |
| *$XS_{i,t}$* | −2.14** | −0.52** | −3.99 | −4.45*** | 0.98* | −2.62** |
| | (−2.00) | (−2.10) | (−1.35) | (−2.61) | (1.65) | (−2.23) |
| *($Rank^*_{it}$-1000)* | −0.27*** | −0.24*** | −0.24*** | −0.18*** | −0.23*** | −0.27*** |
| | (−12.41) | (−10.59) | (−5.88) | (−5.56) | (−10.41) | (−14.34) |
| *($Rank^*_{it}$-1000) × $R2000_{it}$* | 0.21*** | 0.16*** | 0.10 | 0.08* | 0.17*** | 0.18*** |
| | (5.37) | (4.40) | (1.48) | (1.81) | (5.55) | (6.30) |
| *Float $Adj_{i,t}$* | −0.08 | −0.10 | −0.35* | −0.19 | −0.09 | −0.03 |
| | (−0.84) | (−0.94) | (−1.75) | (−1.55) | (−1.16) | (−0.49) |
| Year effects | y | y | y | y | y | y |
| Observations | 13,053 | 8,787 | 5,361 | 10,937 | 17,349 | 14,582 |

This table presents an instrumental variable estimation based on Equations (1) and (2). The second-stage regression presents payout policy variables as a function of instrumented institutional ownership and its interaction with our proxies for agency costs or information asymmetry,

# 9 Cross-sectional evidence: effects are stronger when agency problems are higher (Table 5)

Table 5 estimates (4) for dividends, using alternative agency proxies $XS$.

## 9.1 Table 5

Each column uses a different proxy for agency problems. The coefficient on $\widehat{IO} \times XS$ measures how the marginal effect of IO differs in the high-$XS$ group. The paper's main message is that the overall IO effect on dividends is **driven by firms with higher expected agency costs**.

## 9.2 What the results imply

The evidence indicates a stronger IO→payout effect when:

- the CEO is also the chairman (CEO duality),
- CEO ownership is *lower*,
- board size is larger,
- governance is weaker (higher GIM index),
- profitability is lower (lower ROA),
- cash flows are high and market-to-book is low (free-cash-flow conditions).

This pattern is consistent with Jensen's free cash flow logic: distributing cash can mitigate agency problems, and institutions appear to induce higher payout precisely where the marginal value of monitoring is higher.

# 10 Summary

The paper provides a chain of evidence:

1. Russell index thresholds create a discontinuous shift in index weights (Figure 1), generating a jump in institutional ownership at the cutoff (Figure 2, Table 2).

2. Instrumented increases in institutional ownership causally increase dividends and total payout (Table 3).

3. The payout effect is accompanied by greater shareholder engagement via proposals and voting (Table 4).

4. The effect is strongest in high-agency environments (Table 5), pointing to monitoring/agency-cost mechanisms.

# 11 Conclusion

## 11.1 Summary

This paper concludes that **institutional ownership causally increases corporate payout**, especially **dividends**. Using the annual Russell 1000/2000 reconstitution as a source of quasi-exogenous variation in institutional ownership around the index cutoff, the paper shows that firms just inside the Russell 2000 experience a discrete increase in institutional ownership relative to otherwise similar firms just inside the Russell 1000. Instrumenting institutional ownership with this index-threshold discontinuity, the authors estimate the causal effect of institutional ownership on payout policy.

The core findings form a coherent chain:

(i) **First-stage relevance:** crossing the Russell threshold causes a statistically strong jump in institutional ownership, driven mainly by **quasi-indexers** (and to a lesser extent transient institutions), consistent with index/benchmark demand.

(ii) **Main second-stage result:** higher institutional ownership **raises dividends** (levels and dividend yield) and increases the probability of paying dividends.

(iii) **Broader payout:** institutional ownership also increases **total payout** (dividends + repurchases); repurchases rise as well, though their significance is weaker in the tightest bandwidth and stronger in broader bandwidths.

(iv) **Mechanism evidence:** ownership-induced changes are accompanied by greater shareholder engagement (proposal activity and voting patterns), consistent with a **monitoring/agency** channel rather than pure correlation.

(v) **Heterogeneity consistent with agency:** the effect of institutional ownership on dividends is **stronger when expected agency problems are higher**, consistent with the view that payout is used (and demanded) as a governance device.

## 11.2 Key quantitative conclusions

The paper emphasizes economically meaningful magnitudes for firms around the Russell cutoff:

- The estimates imply that a **one-percentage-point** increase in institutional ownership causes about a **$7 million (roughly 8%)** increase in dividends.
- A **10 percentage-point** increase in institutional ownership raises dividend yield by about **one percentage point** (relative to an average dividend yield of about 2% among comparable Russell 1000 firms) and raises dividend-paying propensity by about **20%** (relative change).

These magnitudes underscore that even ownership changes associated with index tracking can translate into sizable real policy adjustments.

## 11.3 Why higher institutional ownership increases payout

The paper's intuition is fundamentally an **agency/monitoring** logic:

**Step 1: payout is a governance margin.** Dividends (and total payout) reduce free cash flow inside the firm and can limit managerial discretion. When agency problems are larger, the marginal benefit of committing to payout is higher.

**Step 2: institutions raise monitoring and governance pressure.** Institutional investors, even those that are not "activists" in the narrow sense, can exert governance pressure through proxy voting, proposal support, and engagement (directly or via stewardship/proxy-advisory channels). This increases the expected cost to managers of retaining discretionary cash (and/or increases the attractiveness of payout to shareholders).

**Step 3: the Russell threshold isolates ownership changes that are plausibly not chosen by the firm.** Around the cutoff, small differences in market capitalization rank lead to large differences in index weight and index-related demand. Thus, the induced variation in institutional ownership is plausibly orthogonal to firm fundamentals in a narrow neighborhood, allowing the payout response to be interpreted as causal.

**Step 4: why the effect is strongest where agency is most severe.** If the mechanism is governance/agency, then institutional ownership should matter most when internal governance is weaker or the free-cash-flow problem is more pronounced. The paper finds exactly this heterogeneity pattern.

## 11.4 Interpretation of the causal parameter (what the IV identifies)

Because the strategy uses index-induced ownership changes, the 2SLS estimate can be interpreted as a **local average treatment effect** for firms near the Russell 1000/2000 cutoff whose institutional ownership is moved by index assignment, primarily through quasi-indexer and benchmark-driven demand. A key message is that **even "nonactivist" institutional ownership can have first-order effects on corporate policy.**

## 11.5 Takeaway

*Russell index thresholds generate quasi-exogenous jumps in institutional ownership, and instrumented increases in institutional ownership causally raise dividends (and total payout), with stronger effects in high-agency settings and accompanying evidence of greater shareholder governance engagement.*